

THE OMNI-COMPASS UNIFIED GOVERNING CONVERGENCE CONTROL CORE ENGINE

THE OMNI-COMPASS EQUATION FORM

1	$\frac{dE}{dt} = -\alpha_E E + \beta_{\text{int}}(t) + \beta_{\text{ext}}(t) + v_{\text{eff}}$
2	$\frac{dU}{dt} = \mu U(1 - U^2) - \frac{dE/dt}{E_{\text{max}}} - \lambda_U U + u, \quad u \leq U_{\text{AUTHORITY}}$
3	$\frac{dI_U}{dt} = 1 - U - \sigma_1 E - \delta S - \lambda_I I_U$
4	$v_{\text{eff}} = \chi(t) c \tanh[\lambda_0 + \lambda_1(U - U_t) + \lambda_2 S], \quad \chi(t) = \cos(\omega_B t/2)$
5	$\Phi(S) = \frac{1}{2}\alpha_s S^2 + \frac{1}{4}\beta_s S^3 - \delta S$
6	$\partial S = F_{\text{state}}(S) = -\frac{\partial \Phi(S)}{\partial S} = \delta - \alpha_s S - \frac{3}{4}\beta_s S^2$
7	$\partial^2 B + \frac{\omega_B}{Q_B} \partial B + \omega_B^2 B = \gamma_c \delta S \quad [\text{continuous}]$
8	$R_B[n] = D_h^2 B[n] + \frac{\omega_B}{Q_B} D_h^- B[n] + \omega_B^2 B[n] - \gamma_c \delta S[n] = 0 \quad [\text{discrete audit}]$

THE OMNI-COMPASS SYMBOL CHART

E Error / misalignment energy	U Coherence / alignment index	I_U Information / internal-coherence channel	S Coherence-state potential	B Bath field
B_{dot} Bath-field time derivative	α_E E-channel decay / feedback rate	$\beta_{\text{int}}(t)$ Internal forcing function	$\beta_{\text{ext}}(t)$ External forcing function	v_{eff} Bounded effective drive
μ Bistable (double-well) strength	λ_U U-channel damping	u Bounded supervisory command	$U_{\text{AUTHORITY}}$ Command authority bound	E_{max} Per-trajectory error-rate normalization
σ_1 E-coupling into I_U	δ Linear coherence coupling parameter	λ_I I_U damping	$\chi(t)$ Phase multiplier $\cos(\omega_B t/2)$	c Limiting velocity
λ_0 Saturation bias	λ_1 Saturation gain	λ_2 S-state coupling gain	U_t Target coherence state	$\Phi(S)$ Potential-field function
$F_{\text{state}}(S)$ Canonical S-state flow	α_s S-potential quadratic coefficient	β_s S-potential cubic coefficient	ω_B Bath angular frequency	Q_B Bath quality factor
γ_c Bath coupling gain	$R_B[n]$ Discrete bath residual (audit)	D_h^2 Centered second difference	D_h^- Backward first difference	h Microstep size